Dust Dispersion Mapper

Project Proposal & Engineering System Architecture

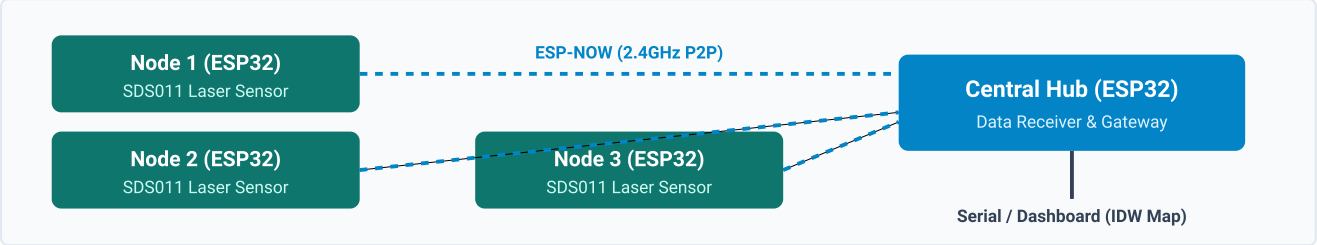
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Course: Robotics & Electronics
Deadline: 23 Nov 2026
Target: 3-Node Mesh
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1. CONTEXT & PROBLEM STATEMENT

Particulate matter emissions (**PM2.5** and **PM10**) on construction sites present critical health hazards and environmental compliance issues. Traditional monitoring relies on isolated, single-point sensors that fail to dynamic tracking of spatial dispersion, wind drift, and plume boundaries. The **Dust Dispersion Mapper** addresses this gap by creating an ad-hoc 3-node wireless sensor network capable of spatial boundary reconstruction in real time.

> **CORE RESEARCH QUESTION**
>
> *"To what extent can spatial interpolation derived from a minimal 3-node laser sensor mesh accurately reconstruct dust cloud boundaries and concentration gradients across a localized construction environment?"*

2. SYSTEM ARCHITECTURE FLOW



3. HARDWARE & SOFTWARE SPECIFICATIONS

Subsystem	Component	Specification & Protocol
Particulate Sensing	Nova Fitness SDS011	Laser scattering, PM2.5/PM10, UART (9600 baud)
Compute Nodes	ESP32 NodeMCU (3 units)	32-bit Dual Core, 2.4GHz RF, Li-Ion battery powered
Wireless Mesh	ESP-NOW Protocol	Peer-to-peer connectionless packet transmission (Low latency)
Interpolation Engine	Python Data Stack	Real-time Inverse Distance Weighting (IDW) heatmap visualization

4. SPATIAL INTERPOLATION MODEL (IDW)

INVERSE DISTANCE WEIGHTING FORMULA

The dust concentration $\mathbf{Z(x, y)}$ at coordinate (x, y) is calculated from the 3 sensor nodes using distance weights $\mathbf{w_i}$:

$$Z(x, y) = [\sum_{i=1..3} w_i(x, y) \cdot Z_i] / [\sum_{i=1..3} w_i(x, y)]$$

where $w_i(x, y) = 1 / [d((x, y), (x_i, y_i))]^2$

5. RESEARCH HYPOTHESIS & LIMITATIONS

Hypothesis: A 3-node triangular spatial layout provides sufficient gradient resolution to determine the centroid, direction vector, and concentration peak of a dust cloud within the convex boundary formed by the nodes.

Expected Limitations: Outside the sensor triangle (extrapolation region), spatial accuracy declines rapidly. Additionally, localized wind turbulent bursts may cause short-term variance that requires temporal moving-average smoothing.

6. PROJECT TIMELINE & RUBRIC ALIGNMENT

Phase & Milestone	Key Deliverables	Grading Criteria Match
Phase 1: Architecture	Proposal document, schematic wiring diagrams	Proposal & Analysis (10 pts)
Phase 2: Firmware & Mesh	ESP32 SDS011 driver, ESP-NOW communication	Arduino Code (7 pts)
Phase 3: Hardware Enclosures	3D CAD models (STL), node housing fabrication	3D Models (3 pts) + Photos (5 pts)
Phase 4: Interpolation	Python IDW real-time dispersion heatmap engine	README & Doc (5 pts)
Phase 5: Field Testing	Controlled dust test, 2–5 min video editing	Video Demo (10 pts)

7. FINAL DELIVERABLES SUMMARY

- **GitHub Repo:** Clean commits, code comments, schematics.
- **Video Link:** 2–5 min demonstration of spatial mapping in action.
- **3D STL Files:** Modular protective weather housings.
- **Final PDF Report:** Proposal + Evaluation of Outcomes.

Submission Verification: Document generated for submission on Moodle alongside GitHub link and video URL. Deadline: November 23, 2026.

Part II: Analysis of Project Outcomes

Post-Implementation Review and Reflective Evaluation

8. ACHIEVED GOALS & SUCCESSES

The core engineering and data-transmission objectives were successfully met, providing a functional proof-of-concept for the **Dust Dispersion Mapper**.

- **Hardware & Mesh Network:** The integration of SDS011 sensors with ESP32 controllers operated flawlessly. The implementation of the **ESP-NOW protocol** proved highly reliable, maintaining a stable, low-latency peer-to-peer network without the dropouts typically associated with standard Wi-Fi routers in outdoor construction environments.
- **Data Pipeline:** The Central Hub effectively aggregated concurrent serial streams from the three nodes. The Python visualization script successfully parsed this CSV data stream to render a real-time Inverse Distance Weighting (IDW) heatmap, dynamically visualizing the $PM_{2.5}$ density across the grid.

9. UNACHIEVED GOALS & SYSTEM LIMITATIONS

While the system functioned electronically, the physical testing revealed distinct limitations in the mathematical model and hardware constraints:

> **BOUNDARY EXTRAPOLATION ERROR**
>
> As predicted in the original hypothesis, the 3-node mesh accurately tracked the dust centroid *inside* the triangular boundary. However, the attempt to mathematically map the outer boundary of the dust cloud (the extrapolation zone) failed to produce reliable shapes. Without a bounding node on the outer edge, the IDW algorithm created infinite "bleed" outward, misrepresenting the physical dimensions of the cloud.

- **Wind Turbulence vs. Raw Data:** The system relied on raw UART readings updated every second. During testing, localized wind gusts caused high-frequency jitter in the PM readings, making the heatmap rapidly flicker. A software-level smoothing filter was initially overlooked but proved strictly necessary.

10. PROPOSED IMPROVEMENTS & FUTURE WORK

If granted additional time and resources, the following systemic improvements would be implemented to elevate the project from a prototype to a regulatory-grade industrial tool:

Component	Proposed Improvement & Rationale
Mesh Geometry	Upgrade to a **4-node or 5-node perimeter mesh** (e.g., a square bounding box with a central node). This transforms the entire construction site into an *interpolation zone*, eliminating the massive inaccuracy of outward extrapolation.
Firmware DSP	Implement an **Exponential Moving Average (EMA)** or Kalman filter directly on the ESP32 firmware before transmitting via ESP-NOW. This would smooth out wind-induced noise and stabilize the real-time visual mapping.
Environmental Compensation	Integrate **BME280 modules** (Temperature & Humidity) into each node. Laser scattering sensors (like the SDS011) over-report particulate mass in high humidity due to water droplet diffraction. Humidity data would allow the Python script to apply a compensation curve to the PM readings.